

Adaptive Immunity I & II

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Session Objectives

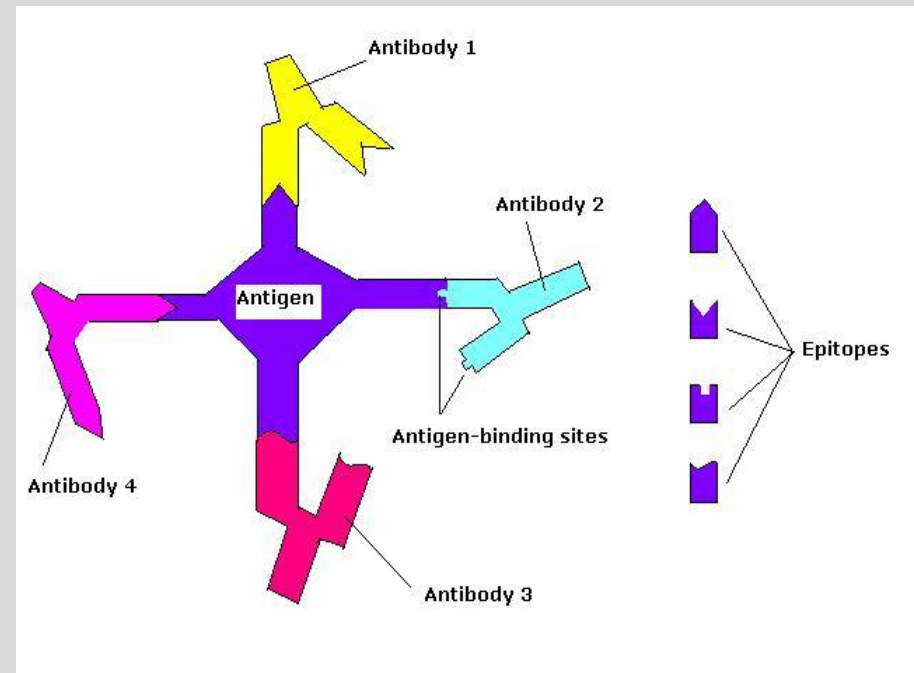
1. Outline the structure of immunoglobulins and their types
2. Explain the role of MHC, TCR & BCR receptor molecules
3. Discuss how different lymphocytes are activated
4. Compare and contrast the four types of hypersensitivity reactions
5. Compare and contrast the humoral and the cell-mediated immune response and the roles of the corresponding key cells (Th1/Th2/Th17)
6. Discuss the regulation of the immune response

Molecules of the adaptive IR (overview)

- Immunoglobulins (BCR)
- MHCs
- TCR
- Cellular interaction:
 - a) Cytokines/Chemokines
 - b) Adhesion molecules
 - c) CD molecules

Igs

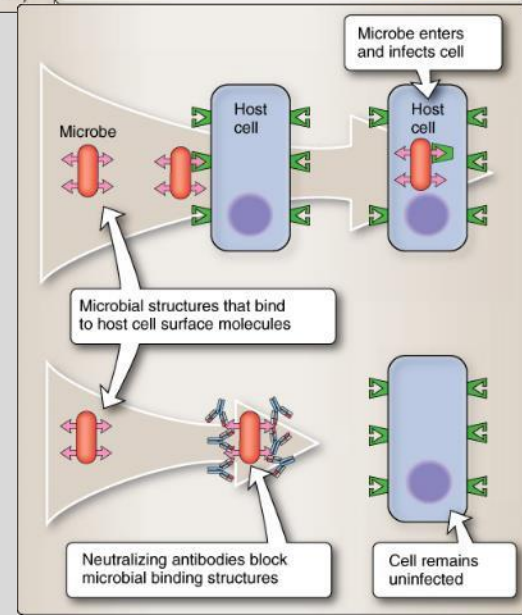
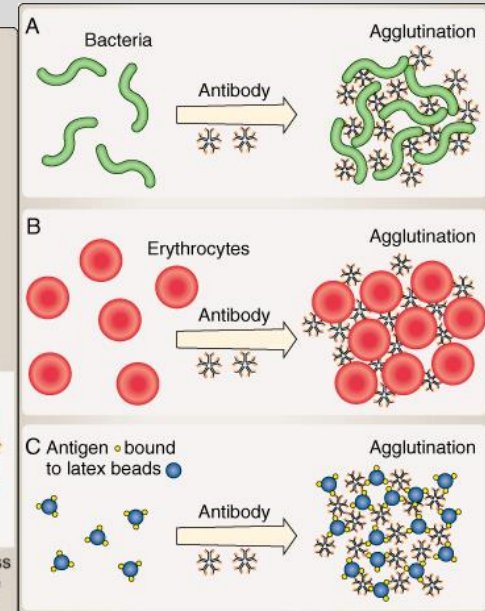
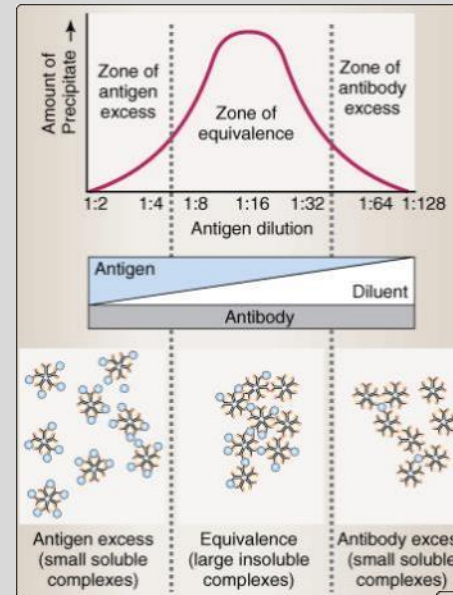
- Ig: soluble (*Humoral*) molecules with specificity to an epitope
- Ag-Ab rxns: Immobilize (cross-link), neutralize, opsonize



https://upload.wikimedia.org/wikibooks/en/c/c8/Chem114A_epitopes.jpg

Ag-Ab reactions

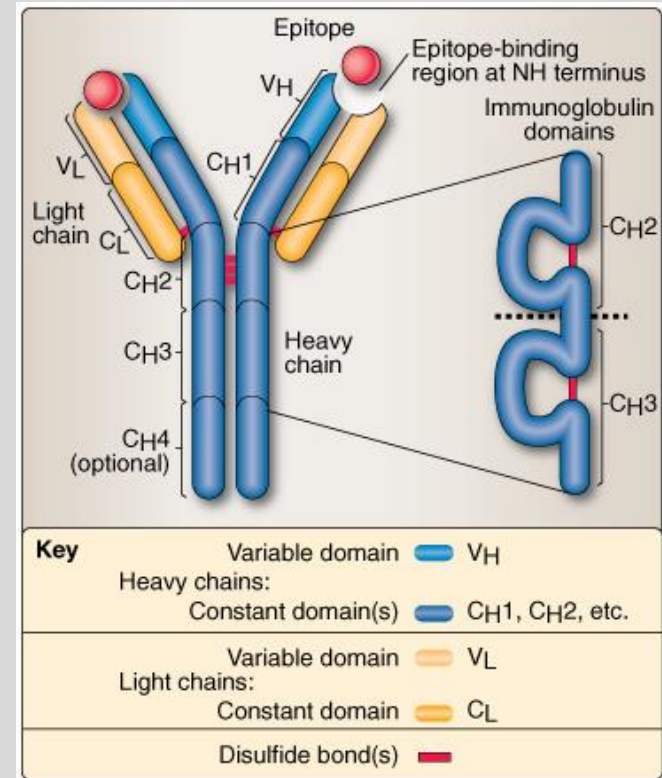
1. The precipitin reaction
2. Agglutination/cross-link (IgM, IgA)
3. Neutralization (IgG, IgA)
4. Opsonization (phagocytosis)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Igs

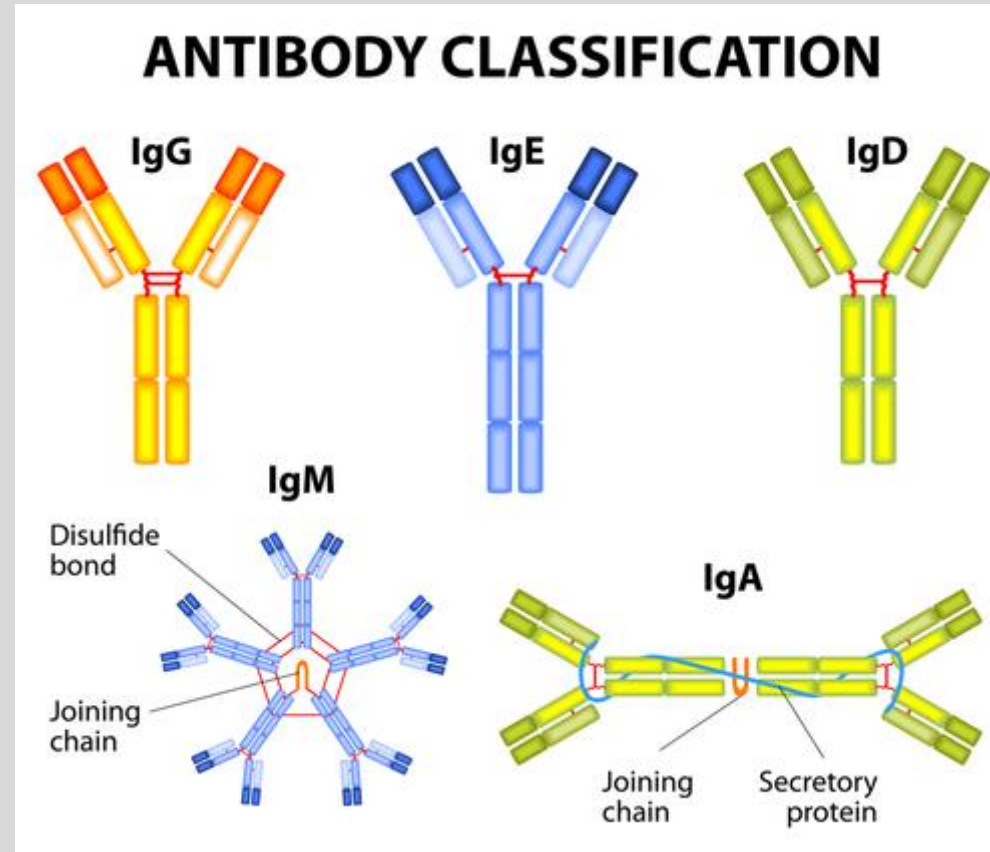
- 4 polypeptides: 2 heavy & 2 light chains linked by S-S bonds
- 5 types of heavy chains: μ , γ , α , δ , ϵ (isotypes)
- Papain digestion:
 - a) Fab (Ag-binding fragment)
 - b) Fc (constant)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Ig isotypes

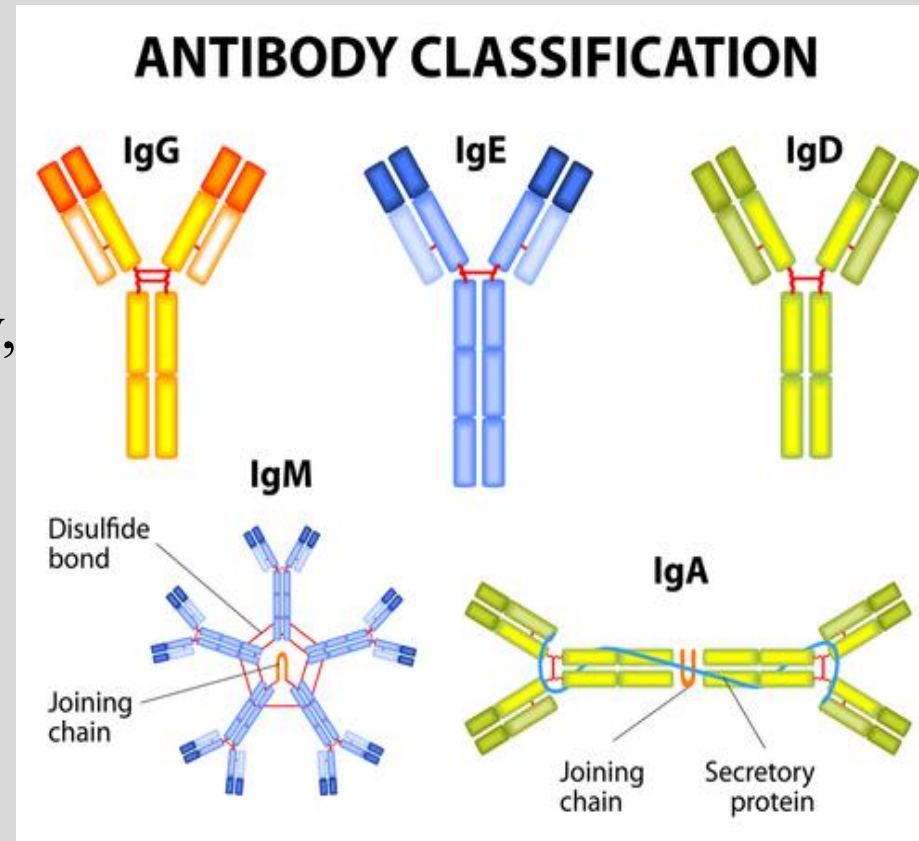
- IgM (μ):
 - a) Pentamer (secreted)
 - b) Monomer (B-cell surface)
 - c) 1st Ab
 - d) Agglutination (immobilize)
 - e) Complement fixation
 - f) Doesn't cross placenta
- IgD (δ): monomeric; BCR



<https://aws.labome.com/figure/te-152-2.png>

Ig isotypes

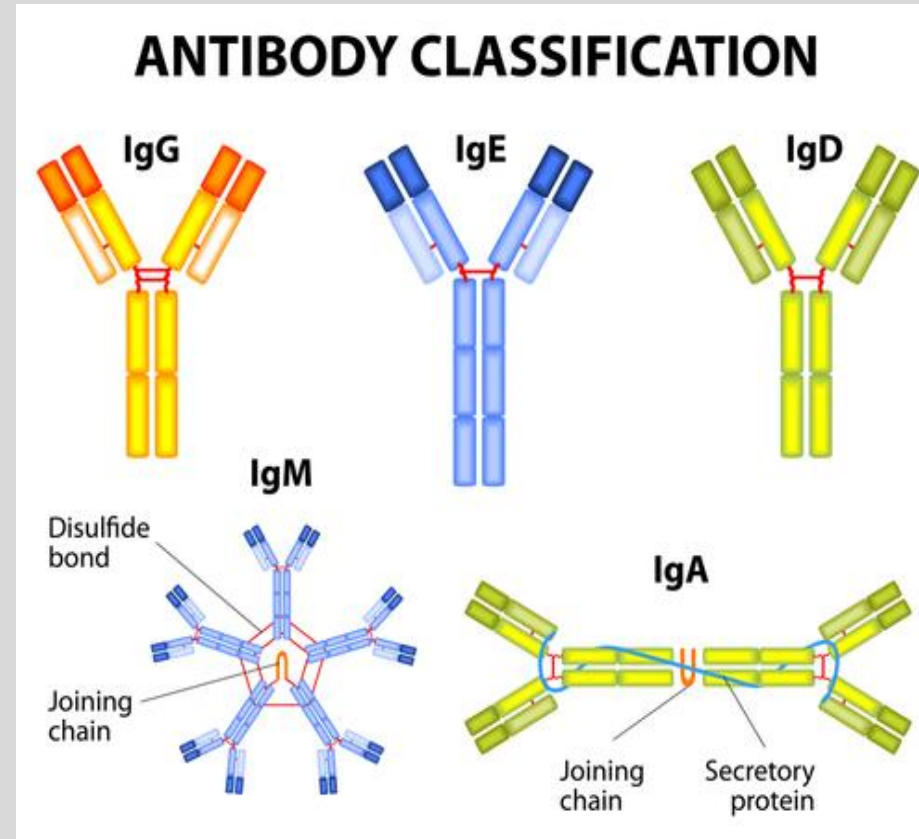
- **IgG (γ):**
 - a) Most common in serum
 - b) Monomer (surface & secreted)
 - c) 2nd Ab released
 - d) Neutralize, opsonize, complement, hypersensitivity, Ab-dependent cell-mediated cytotoxicity
 - e) Crosses placenta
- **IgA (α):**
 - a) Monomer (serum)
 - b) Dimer (secretions: Tears, saliva, milk, etc.)
 - c) Most Ab produced daily
 - d) Mucosal I (MALT)



<https://aws.labome.com/figure/te-152-2.png>

Ig isotypes

- IgE (ϵ) :
 - a) Lowest in serum
 - b) Monomer
 - c) Adsorbed on mast cells & eosinophils (worms)
 - d) Immediate hypersensitivity (Type I)



<https://aws.labome.com/figure/te-152-2.png>

The constant regions in the heavy chain of antibodies dictate the molecule's

Epitope

0%

Fab fragment

0%

Variable domain

0%

Isotype

0%

Which of the following Abs crosses the placenta and provide passive immunity to the fetus?

IgA

0%

IgM

0%

IgE

0%

IgG

0%

The epitope-binding site on the antibody is located ____

on the Fab fragment

0%

on the Fc fragment

0%

on the constant region

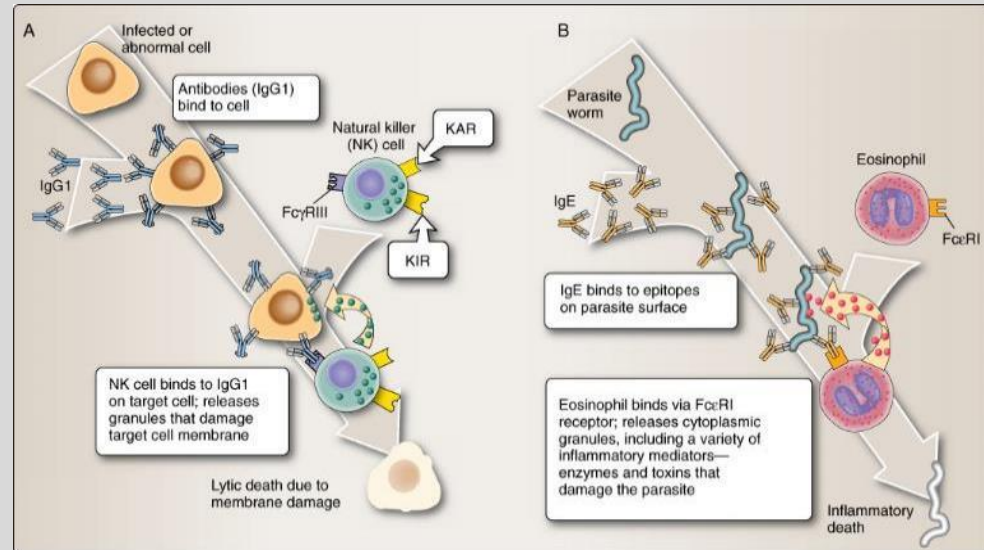
0%

on the variable region

0%

Ag-Ab reactions

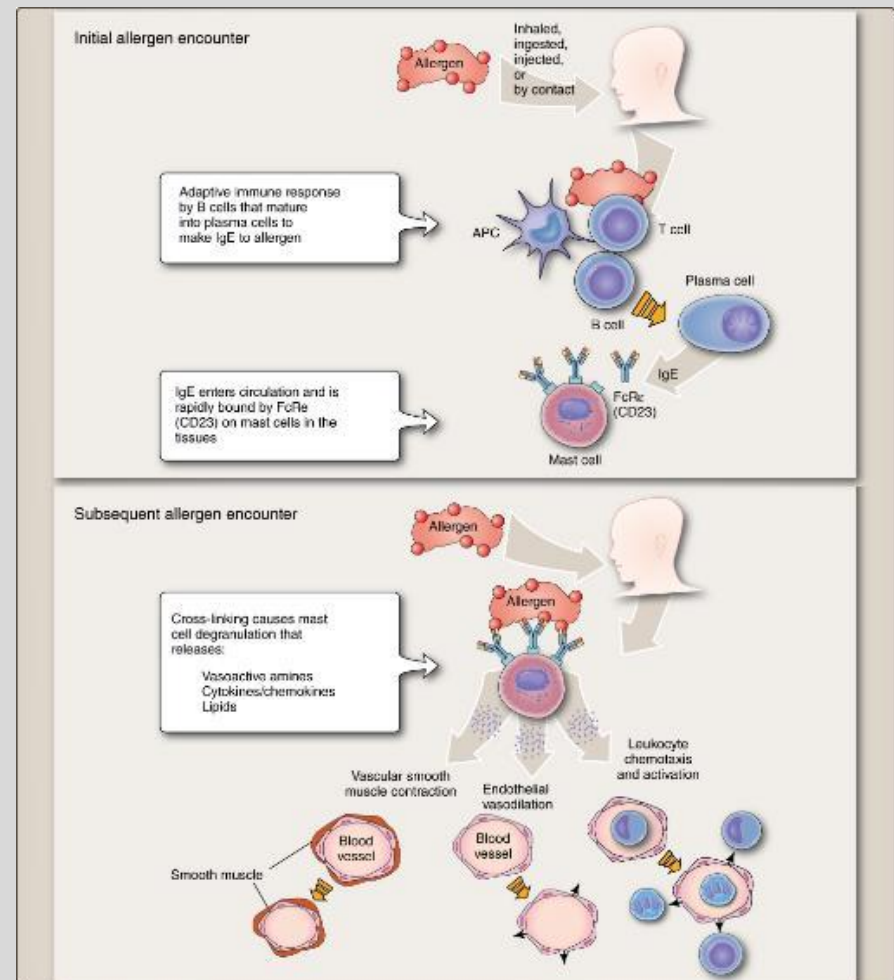
- Antibody-dependent cell-mediated cytotoxicity:
 - Ab-tagged bacteria, parasites or worms attract NK cells & eosinophils



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

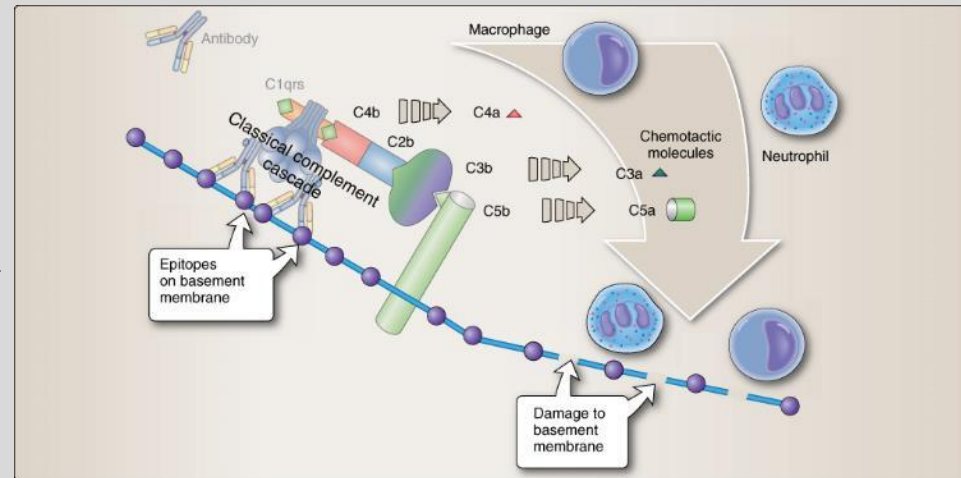
Hypersensitivities (Ab-mediated)

- Type I (Immediate hypersensitivity):
 - a) IgE → mast cells & basophils
 - b) Asthma/allergies, anaphylaxis

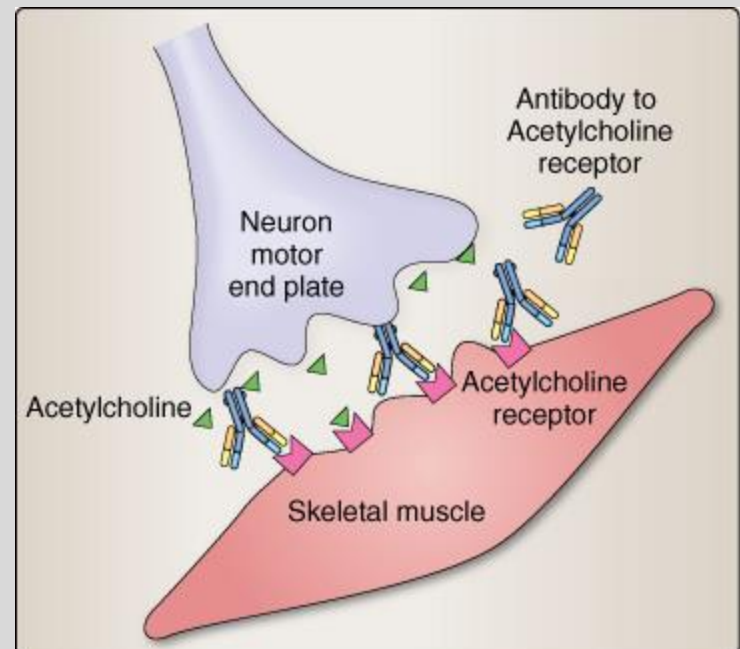


Hypersensitivities (Ab-mediated)

- Type II hypersensitivity (Antibody-dependent cell-mediated cytotoxicity):
 - IgM/IgG interaction with cell membranes or extracellular matrix
 - Complement, NK cells involved
 - Hemolytic anemia from mismatch
 - Bind to basement membrane → inflammation and tissue injury
 - Ab blocks function (Grave's disease & Myasthenia Gravis); Rheumatic fever

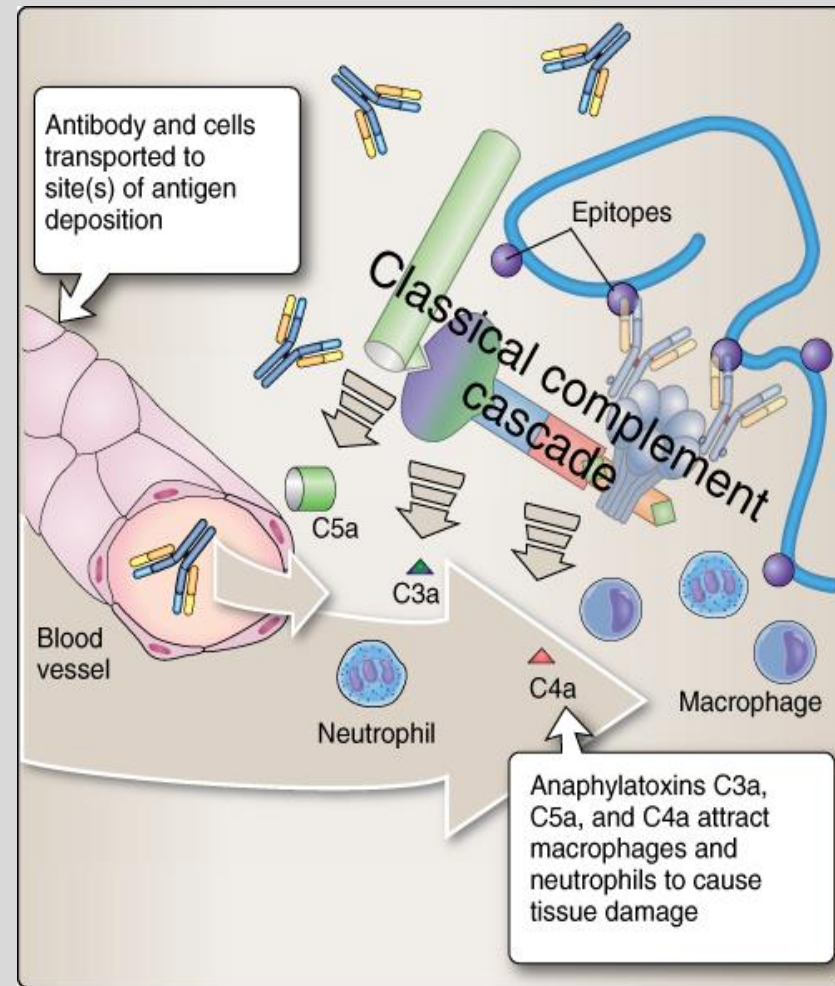


Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher



Hypersensitivities (Ab-mediated)

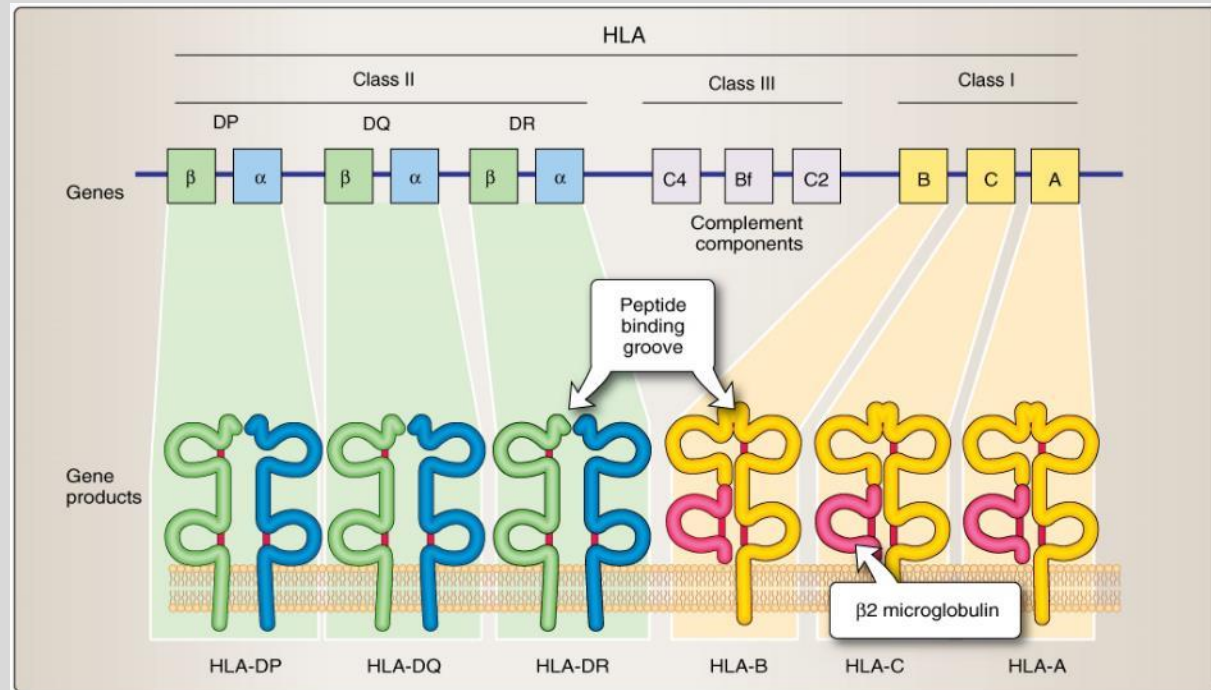
- Type III hypersensitivity (immune complex):
 - Circulating Ag-Ab complexes (IgG) → vasculitis
 - Localized (Arthus): abscess or necrotizing vasculitis
 - Systemic (serum sickness):
 - a) Exogenous: non-human Abs/drug induced (e.g. penicillin)
 - b) Endogenous: e.g. SLE



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

MHC molecules

- MHC=HLA
- 3 classes:
 - a) MHC I
 - b) MHC II, &
 - c) MHC III
(*complement components*)



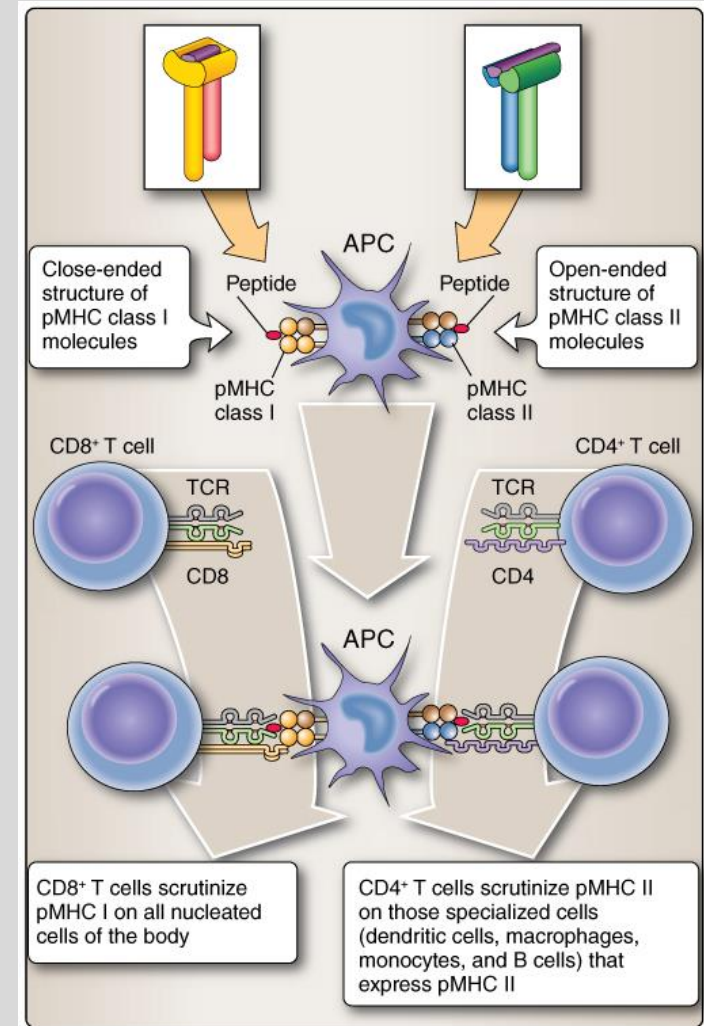
Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

MHC molecules

- MHC I:
 - a) **All nucleated cells**
 - b) Function: self reporting (viral-infected & tumors)
 - c) HLA-A, B & C (100 alleles at each locus)
- MHC II:
 - a) Expressed mostly on APCs (DC, Macrophages)
 - b) But also B-cells; some activated T-cells, some specialized epithelial cells in the thymus & intestine)
 - c) Function: involved in antigen presentation

TCR

- Heterodimer: $\alpha\beta$ or $\gamma\delta$
- Associate with CD3 (signal transduction)
- MHC-restricted: must bind to peptide fragments associated with MHC I & II (*stabilized by either CD4/CD8*)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Cellular interaction

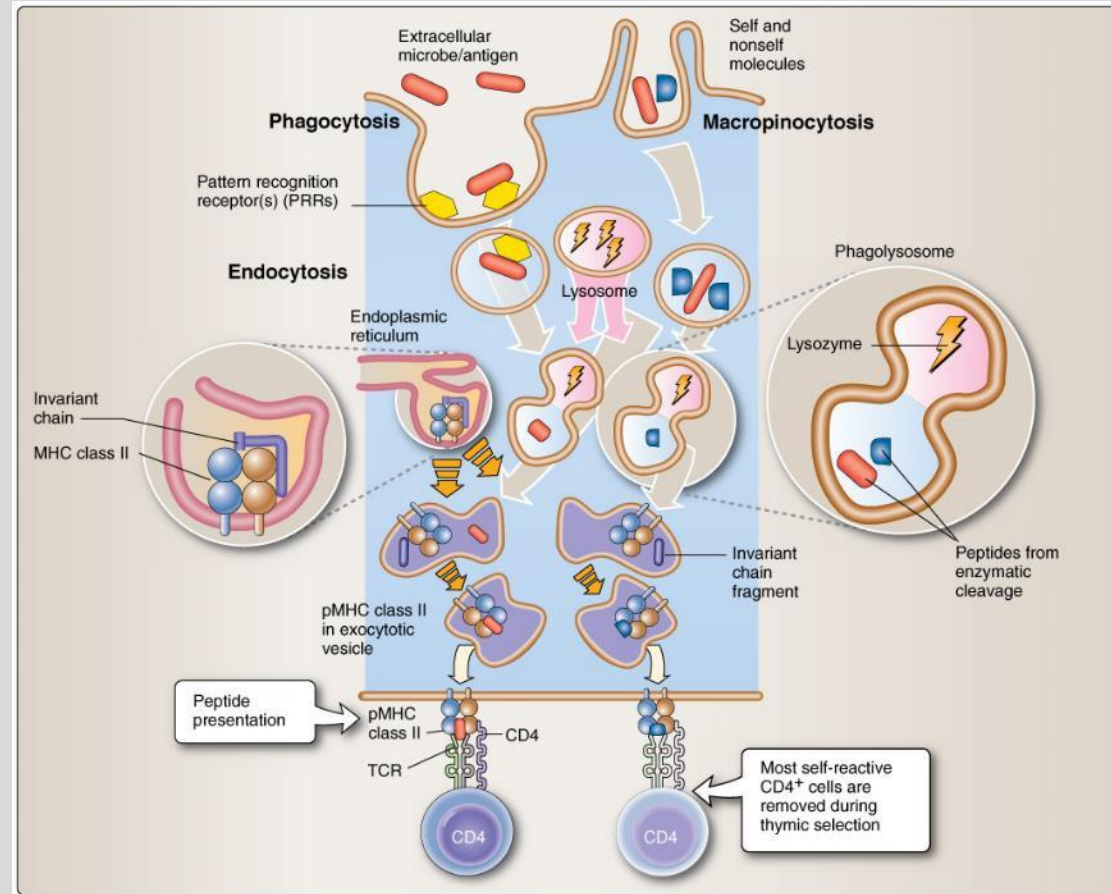
- Cytokines: low MW soluble messengers
- Chemokines: chemoattractant cytokines
- Adhesion molecules: integrins & selectins
- CD:
 - a) Indicator of function
 - b) >350 identified
 - c) CD3, CD4 & CD8

Lymphocyte activation

- Time consuming event with lots of checks & balances
- Main cell of the adaptive IR: T-helper cell
- Antigen presentation (MHC I or II) required for lymphocyte activation (TCR is MHC-restricted)

MHC II presentation

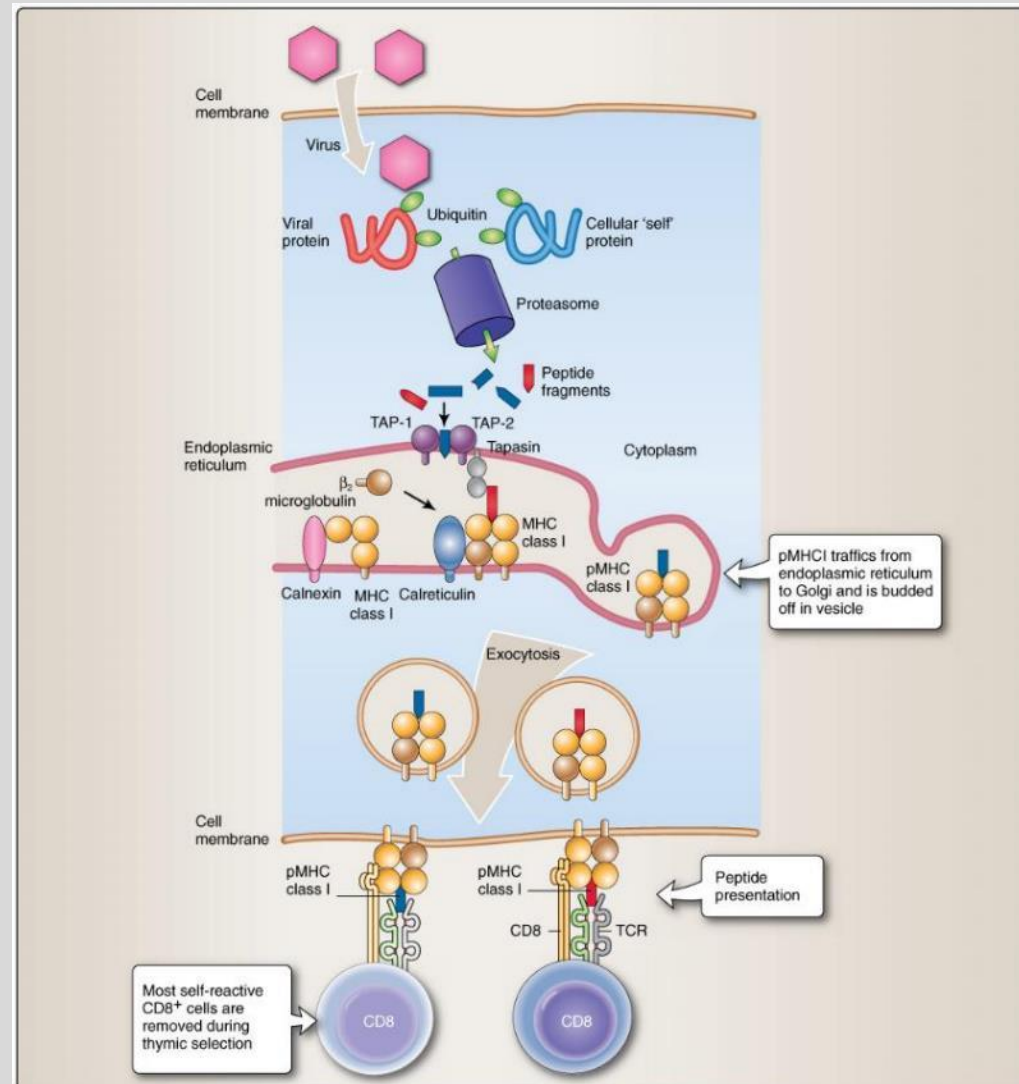
- DC main APC (sentinels)
- Actin-dependent phagocytosis
- PRRs engage PAMPs (extracellular Ags)
- Ab or C-tagged Ags
- Ag displayed on MHC II
- MHC II displays self & non-self Ag (tolerance)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

MHC I presentation

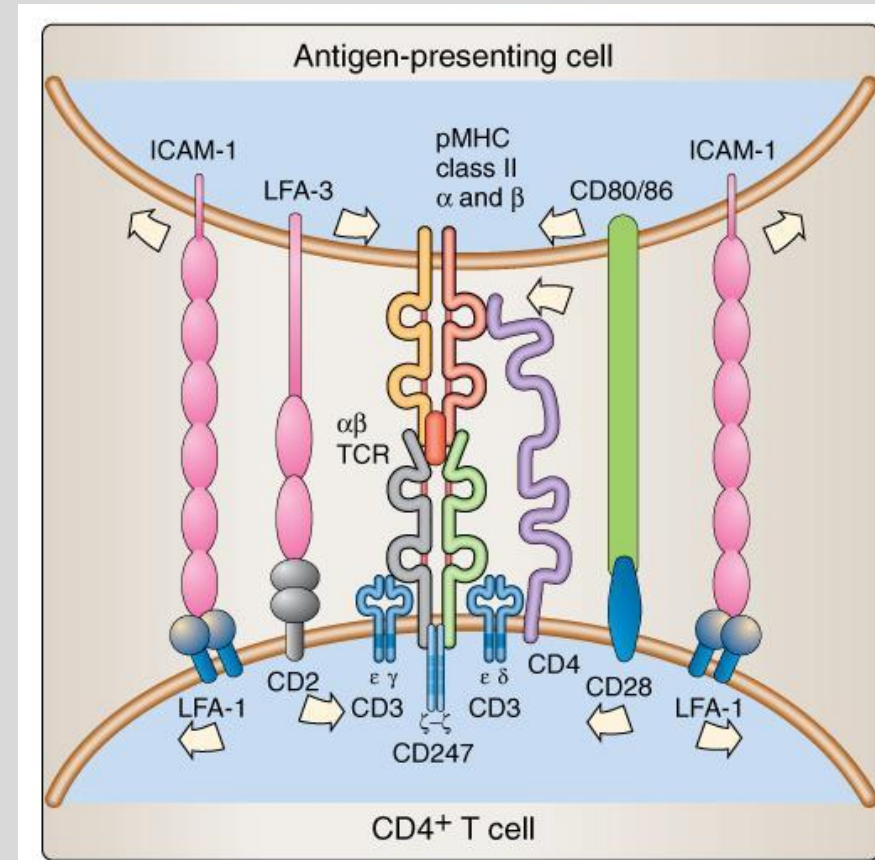
- Intracellular Ags tagged with ubiquitin → proteasome destruction → ER → Golgi → display on MHC I → recognition by CD8+ cells



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

T-helper cell activation (CD4⁺)

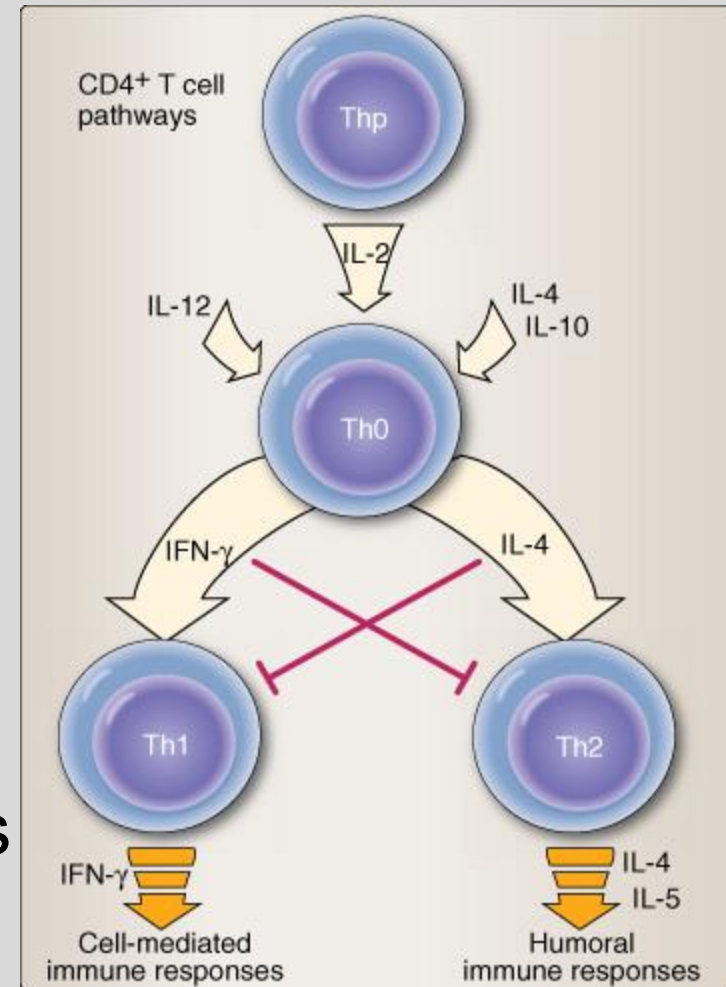
- Only recognize MHC II-associated Ags
- Immunologic synapse: interface between naïve Th & APC
- **Stabilized by CD28** (*costimulatory*) → signal transduction
- Absence of costimulatory signal = anergy/apoptosis



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

T-helper cell activation (CD4+)

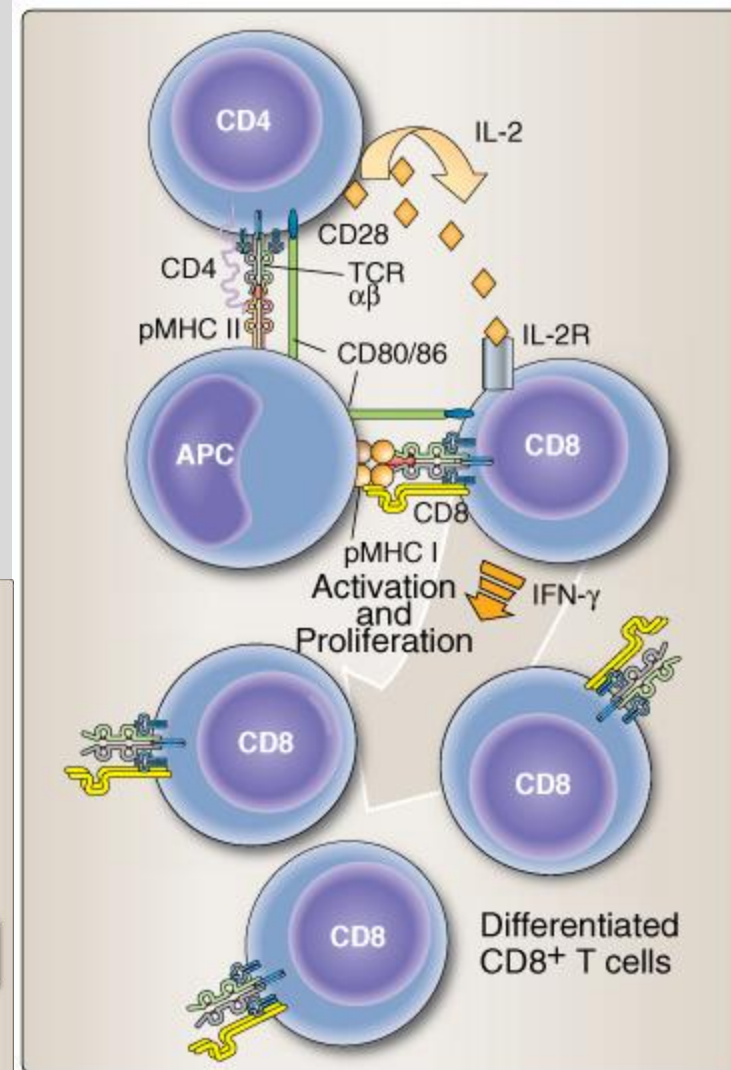
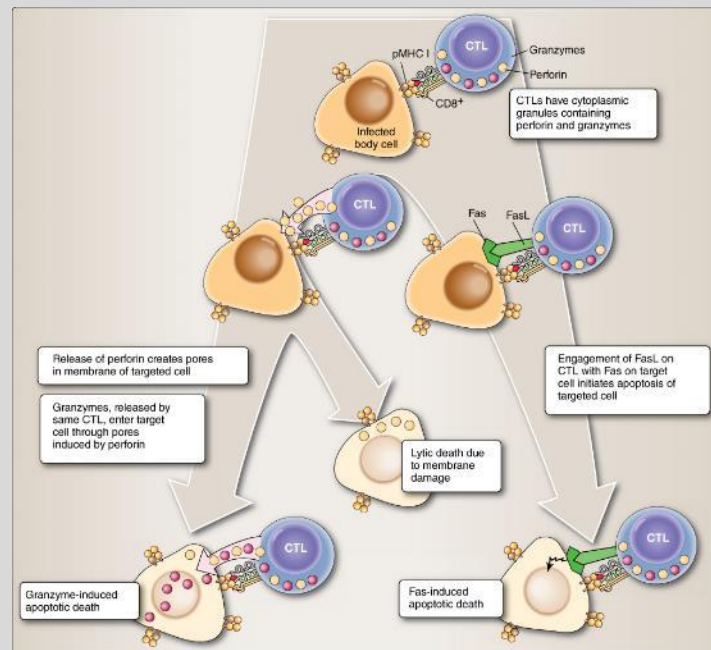
- 1st contact with Ag → prime T-helper
- T-helper → Th1, Th2 or Th17 (*cytokines from APC*)
 1. Th1: IFN- γ /IL-12
 2. Th2: IL-4/IL-10
 3. Th17: IL-17/IL-6/TGF- β
- Some become memory T-cells (*high expression of CD28*)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

T-cell activation (CD8+)

- Recognize Ags on MHC I → expression of IL2R
- With help from Th1 cells → T-cytotoxic
- Perforins & granzymes



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Receptors on T-cells are typically associated with?

CD55

0%

CD3

0%

IL-4

0%

Ubiquitin

0%

B-cell activation

- Recognize Ags in association with MHC class II or without
- Ag binds to BCR → endocytosis → Ag presentation on MHC to CD4+ (T-dependent Ags) → Th2/Th17 (IL-4/IL-6) → plasma cells (some memory)
- T-independent (Ti) Ags (*activate B-cells without T-cells help*)
 - a) Ti-1 Ags: polyclonal activators bind to receptors other than BCR (E.g. LPS); B-cell mitogens
 - b) Ti-2 Ags: multivalent polysaccharides; stimulate only mature B-cells

Memory B-cells undergo point mutations in the coding regions of both light and heavy chains when re-exposed to an antigen. This result in

New Ag specificity

0%

Isotype switching (IgM to IgG)

0%

DNA chromosomal rearrangement to alter specificity

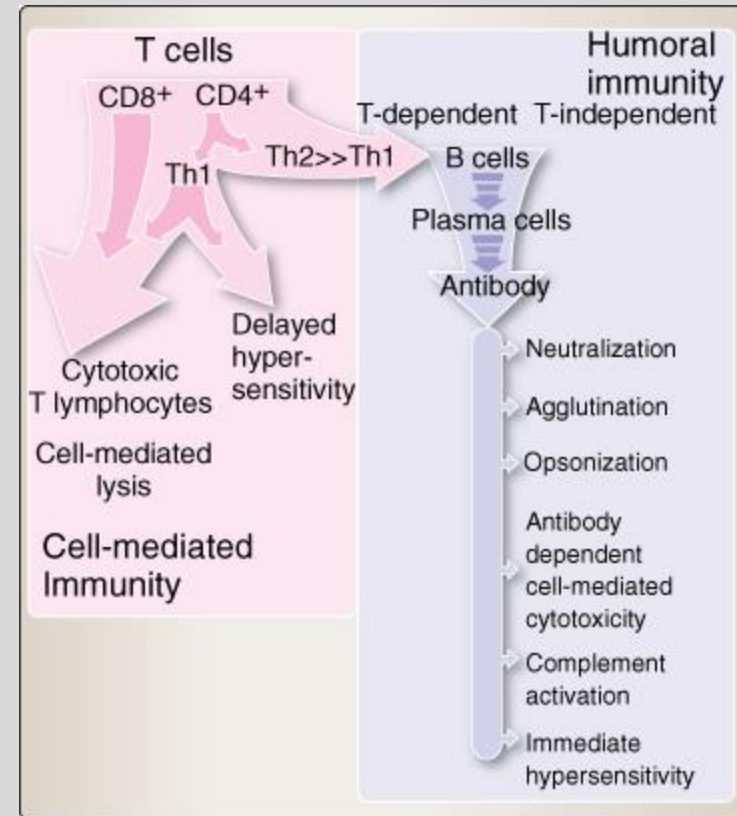
0%

generation of antibodies with increase affinity for its epitope

0%

Two arms of the Adaptive IR

- Humoral IR:
 - a) Abs
 - b) Extracellular pathogens
 - c) Type I, II & III hypersensitivities
- CMI:
 - a) Intracellular pathogens
 - b) Type IV hypersensitivities



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Which of the following is initiated by the interaction of host cell membranes and IgM/IgG but never IgE?

Arthus reaction

0%

Type I hypersensitivity

0%

Type II hypersensitivity

0%

Type III hypersensitivity

0%

The tuberculin skin test is an example of?

Type I hypersensitivity

0%

Type II hypersensitivity

0%

Type III hypersensitivity

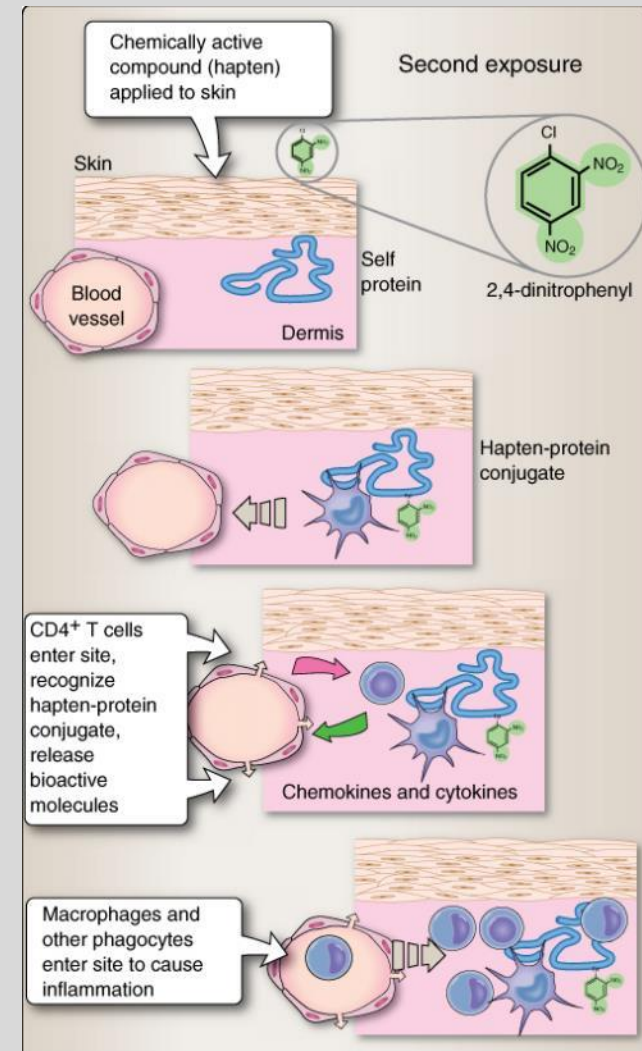
0%

Type IV hypersensitivity

0%

Type IV hypersensitivity

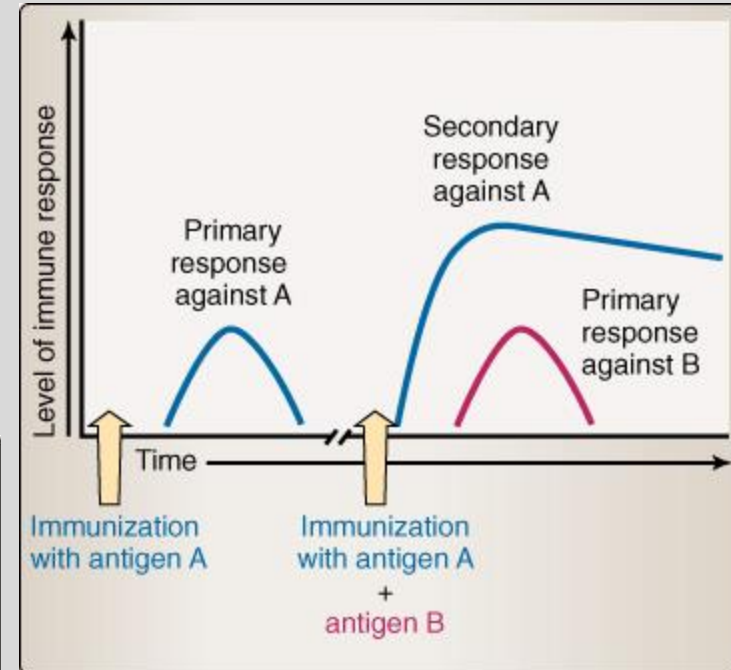
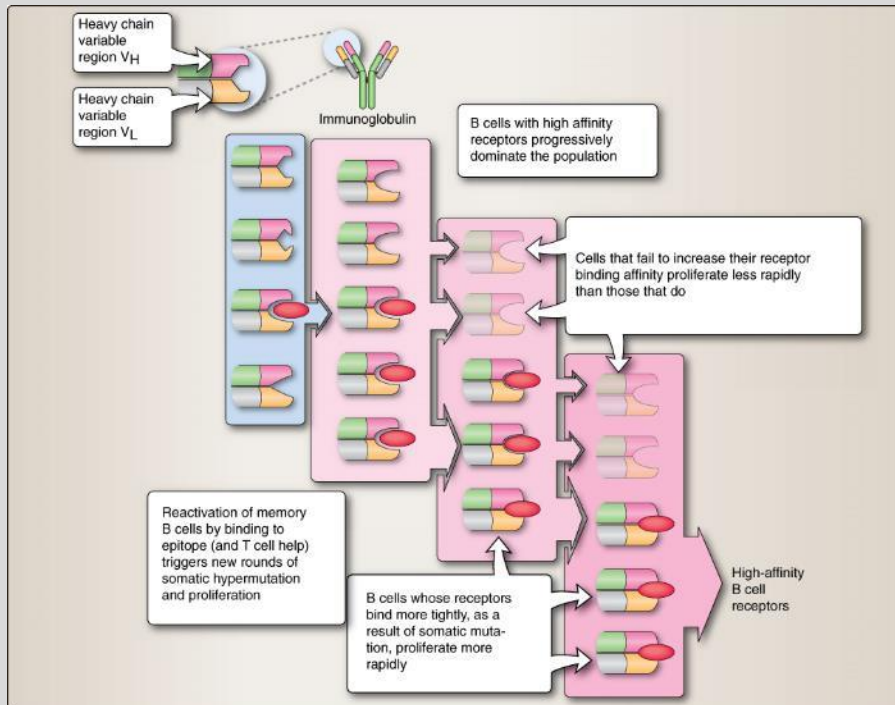
- Th1/Macrophages induced inflammation
 - Topical: contact dermatitis (poison Ivy, TNT, nickel)
 - Ag injected or DTH (48 to 72 hours): due to microbes (*Mycobacteria*, *Leishmania*, *Candida*, mumps virus, many fungal skin tests)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Immunologic memory

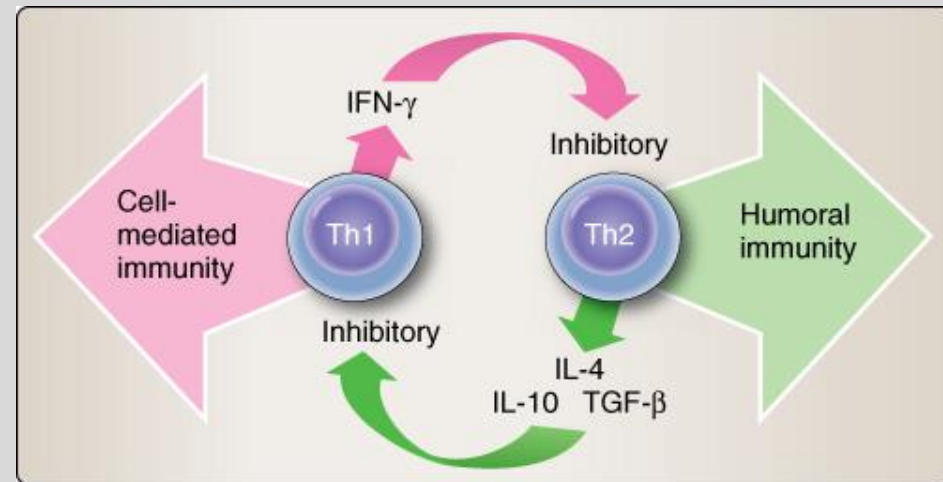
- 1' vs 2' IR
- Isotype switching
- Affinity maturation (hypermutations)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Immune regulation

- Tolerance:
 - a) Anergy (CD28/B7)
 - b) T_{reg} (CD4, CD25 & FOXP3)
 - c) CD8+ suppressor cells (graft rejection; autoimmune diseases)
- Proinflammatory: Th17 (*antagonists of IL-17 very effective in psoriasis*)
- Th1/Th2 paradigm



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Adaptive responses

Adaptive Responses Involved in Clearance of Active Infections

Organisms	IgM, IgG, IgA				IgE	CTL	DTH
	Complement Activation	Opsonization	ADCC	Neutralizing Antibody			
Viruses							
Bacteria (intracellular)							
Bacteria (extracellular)							
Protozoa (intracellular)							
Protozoa (extracellular)							
Fungi							
Flatworms							
Roundworms							

Which of the following cells is implicated on the prevention of autoimmune responses?

APC

0%

Anergic T-cells

0%

DC

0%

Treg

0%

Case study

A 4-year-old female was brought to the ED with dyspnea and uncontrolled muscle spasms. On physical examination the patient had difficulty opening her mouth as well as swallowing. History indicates that the child has *not* received standard childhood vaccination. The patient was recently stung by a bee while playing in the school yard. The physician orders laboratory tests that include microscopic analysis as well as anaerobic culture. Microscopic analysis of the Gram stain is shown below.



Live PollEv Q & A will take place during applied learning sessions

Summary Slide

- Immunoglobulin structure, function, isotypes and involvement in hypersensitivities
- MHC, TCR & BCR structure and function
- Lymphocyte activation (T and B-cells)
- Humoral & CMI (Th1/Th2/Th17)
- Immune memory & regulation
- Clinical case

References

- Core References:

- Abbas, A. K., A. H. Lichtman, and S. Pillai. 2016. 5th ed. **Basic Immunology: Functions and Disorders of the Immune System**. Elsevier Inc.
- Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2022. **Lippincott's Illustrated Reviews: Immunology**, 3rd edition. Lippincott Williams & Wilkins publisher.
- Tao, LE, V. Bhushan and M. Sochat. 2025. **First Aid for the USMLE Step 1**; McGraw Hill Publication.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>